Arm® Cortex®-M55
32-bit Microcontroller

NuMicro® Family
M55M1 Series BSP
Revision History

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Revision 3.01.005 (Released 2026-07-24)

- Add SBOM.
- CMSIS upgrades to 6.3.0 and adds CMSIS-Compiler 2.2.0.
- Startup
  - Reset_Handler_PreInit fixes incorrect timeout check.
- Standard driver
  - FMC: FMC_WriteMultiple supports resume for timeout recovery.
  - I2S: I2S_ORDER_AT_LSB and I2S_ORDER_AT_MSB fix wrong definition.
  - LPTPWM: LPTPWM_ENABLE_OUTPUT fixes disable functionality.
- CMSIS driver
  - USBD: Support HSUSBD DMA transfer.
- Library
  - UsbHostLib supports UAC 2.0.
- ThirdParty
  - FatFs: Upgrade to R0.16.
- Sample code
  - All sample code projects add no-unaligned-access compiler flag.
  - DMIC_I2S_Play and DMIC_WAVRecorder: Add 10 ms DMIC turn-on time.
  - EPWM: EPWM_EnableFaultBrake removes support of multiple brake sources.
  - FMC_XOM, XOMLib, and XOMLibDemo: Refactor for dual bank.
  - HSUSBD mass-storage sample codes: Use memory buffer to replace fixed address.
  - HSUSBD_MassStorage_ScatterGather: Align buffer for Cache/DMA coherence.
  - HSUSBD_RNDIS: Improve performance of bulk transfer with DMA.
  - HSUSBH_USBH_AudioClass, HSUSBH_USBH_UAC_HID, and HSUSBH_USBH_UAC_Loopback: Support UAC 2.0.
  - I2S_Codec_PDMA: Support 24-bit data width.
  - ISP_DFU: Remove dfu-util.
  - Jpeg_Display: Refine memory usage to display larger pictures.
  - LwIP_MQTT: Fix issues of semaphore double-take and local stack buffer.
  - SDH_FATFS: Improve SD card hot-plug response time.
- Tool
  - HIDTransferTest
    - Move from USBD sample codes.
    - Add Control transfer.
  - mcuboot: Upgrade to v2.4.0 with imgtool.

Revision 3.01.004 (Released 2026-03-26)

- Peripheral register
  - fmc_reg.h: Define FMC_ISPSTS_INTFLAG_Pos based on secure state.
- Standard driver
  - EADC: EADC_Open removes EADC_ Calibration call.
  - EPWM: EPWM_EnableFaultBrake supports multiple brake sources.
  - I3C: Update for new I3C sample codes.
  - PMC: PMC_SetPowerDownMode addes NPD3 power-down mode limitation.
  - QSPI: QSPI_GET_TX_FIFO_COUNT adds.
  - RNG: Remove SysTick dependency.
  - SPIM: Improve stability of quad I/O transfer mode and 4-byte address mode.
- CMSIS driver

- Flash: Support PDMA and non-blocking mode operations.
- Library
  - UsbHostLib
    - Support still image capture.
    - Reduce memory usage and fix device disconnect flow.
  - xmodem: Fix 128-byte unalignment send issue.
- Sample code
  - CANFD_CAN_TxRx, CANFD_CAN_TxRxINT, CANFD_CANFD_TxRx, CANFD_CANFD_TxRxINT: Add bus-off recovery handling.
  - CANFD_TxRxINT: Fix wrong bitmask in CANFD00_IRQHandler.
  - EADC samples: Fit max 90 MHz EADC clock frequency.
  - HSUSBH_USBH_UVC: Support video still image capture.
  - I3C_Controller: Replace samples of I3C_Master_IBI, I3C_MasterRW, I3C_MasterRW_PDMA and I3C_SecondaryMaster.
  - I3C_Target: Replace samples of I3C_Slave_HotJoin, I3C_Slave_IBI, I3C_Slave_Wakeup, I3C_SlaveRW, I3C_SlaveRW_PDMA and I3C_SecondaryMaster.
  - ISP samples
    - Fix flash select setting in flash download of Keil project.
    - Fix the issue of “Reset and Run” command not booted from APROM.
  - Jpeg_Display: Add.
  - LwIP_MQTT: Renew flespi token.
  - NuvotonAudioGUI_Firmware: Add.
  - QSPI_DualMode_Flash, QSPI_QualMode_Flash, SPI_Flash: Improve performance.
  - RNG_EntropyPoll, RNG_Random: Remove SysTick dependency.
  - SPIM samples: Update trim procedure to reduce times to erase or program Flash.

Revision 3.01.003 (Released 2025-10-29)

- Peripheral register
  - eadc_reg.h: Remove some redundant bit-field macros.
- Standard driver
  - retarget.c: Improve compatibility with CMSIS RTE.
  - CLK: Remove CLK_APLLCTL_APLLSRC_HXT_DIV2.
  - LPPDMA: Redefine LPPDMA_INT_TRANS_DONE and LPPDMA_INT_TEMPTY to support multi-bit mask.
  - PDMA
    - PDMA_INT_TRANS_DONE, PDMA_INT_TEMPTY, PDMA_INT_TIMEOUT, PDMA_DisableInt, PDMA_EnableInt: Support multi-bit mask.
    - PDMA_INT_ALIGN: Add.
  - SPIM: Remove voltage raise for HyperBus device and octal SPI flash.
- CMSIS driver
  - RTE_Device.h: Improve compatibility with CMSIS packs.
  - I2C: Master addes I3C IP support.
  - USBD: Add HSUSBD IP support.
  - VIO: Add NuMaker board support.
- Library

- UsbHostLib: Improve robustness of memory management and EHCI interrupt handling.

■ Sample code
- DMIC_I2S_Play, I2S_Codec_PDMA, I2S_MP3Player, I2S_WAVEPlaye: Fix cache line alignment issues of I2S PDMA buffers.
- EADC_AverageCMP: Add.
- MP3_Recorder: Add USB storage device support.
- VS Code projects upgrade to use GCC 14.3.1.

Revision 3.01.002 (Released 2025-8-8)

■ Peripheral register
- canfd_reg.h
  - CANFD_RXF0A_F0AI_Msk: Replace CANFD_RXF0A_F0A_Msk.
  - CANFD_RXF0A_F0AI_Pos: Replace CANFD_RXF0A_F0A_Pos.
- hsusbd_reg.h: Fix definitions of HSUSBD_EPBUFSTART_SADDR_Msk and HSUSBD_EPBUFSTART_SADDR_Pos.

■ Startup
- SysTick vector on flash uses Default_Handler.

■ Standard driver
- CANFD: Remove unsupported UTSU function.
- HSUSBD: Add HSUSBD_DISABLE_HS_HANDSHAKE and HSUSBD_ENABLE_HS_HANDSHAKE.
- SPI: Add SPI_DISABLE_3WIRE_MODE and SPI_ENABLE_3WIRE_MODE.
- SPIM
  - Raise voltage to 1.2V for HyperBus device and octal SPI flash.
  - Support Infineon octal SPI flash.
- SYS
  - SYS_GPC_MFP0_PC2MFP_UTCPD0_CCDB2: Replace SYS_GPC_MFP0_PC2MFP_UTCPD0_CCDB1.
  - SYS_GPC_MFP0_PC3MFP_UTCPD0_CCDB1: Replace SYS_GPC_MFP0_PC3MFP_UTCPD0_CCDB2.
  - SYS_ResetModule: Add SPIM clock setting handling.

■ CMSIS driver
- CAN, ETH_MAC, Flash, GPIO, I2C, MCI, SAI, SPI, USART, USBD, USBH: Add.

■ Library
- CryptoAccelerator: Improve cache coherence.
- JpegAcceleratorLib: Improve performance.
- UsbHostLib: Add UVC support.

■ Sample code
- AWF_GSensor_Wakeup: Use I/O buffer to replace fixed address.
- DMIC_VAD_Wakeup: Increase VAD power threshold.
- HSUSBD_Audio20_Codec and HSUSBD_Audio20_Headset: Fix no sound issue.
- HSUSBD_HID_Mouse: Improve LPM stability.
- HSUSBD_RNDIS: Add.
- HSUSBD_Video_CAM: Fix UVC image noise issue and race condition of UVC functions.

- HSUSBH_USBH_UVC, HSUSBH_USBH_VCOM_MassStorage and ImageClassification: Add.
- I3C samples: I3C pins enable Schmitt trigger.
- ISP_DFU_20 and ISP_HID_20: Fix USB high-speed issue.
- ISP_I2C and ISP_SPI: Fix potential command failure.
- KeywordSpotting: Add UAC support.
- KS_AESKey: Improve cache coherence.
- ObjectDetection_FreeRTOS, VisualWakeWords: Fix I/O buffer alignment and race condition of UVC functions.
- RTC_TimeAndTick: Remove because almost the same as RTC_Time_Display.
- SecureApplication samples: VS Code projects require Python 3.12 at least for post-build.
- SPIM_HYPER_ExeInHRAM: Fix I/O buffer alignment and GCC project execution failure.
- SPIM_HYPER_RW_MemMap: Fix I/O buffer alignment.
- USBD samples: Improve stability and performance.
- VS Code projects update for Nuvoton NuMicro Cortex-M Pack.

Revision 3.01.001 (Released 2025-2-14)

- Initial release for M55M1.

Revision 3.00.001 (Released 2024-1-31)

- Initial release for M55M1ES.

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